

HydroElasticFEM.jl: A Julia package for the finite element analysis of hydroelastic wave-structure interaction problems

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Software

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Summary

Very large floating structures (VLFS) are gaining attention in many applications, for example in offshore wind platforms, floating breakwaters, flexible wave energy converters, and futuristic floating cities. The interaction of VLFS with free surface waves induces structural deformation and structural stresses that are critical for their performance. Predicting the *hydroelastic* response of VLFS in waves requires a coupled model: the fluid pressure drives structural motion, and structural motion affects the wave propagation through fluid boundary conditions. Therefore, the two systems must be solved together. Traditional approaches treat the fluid and structure as separate sub-problems, exchanging forces and displacements in a staggered loop, which might introduce splitting errors and limit achievable accuracy for strongly coupled configurations. In addition, partitioned schemes suffer to converge when the added mass effect is large, which is particularly relevant for floating flexible thin structures.

HydroElasticFEM.jl is an open-source Julia package (Bezanson et al., 2017) for finite element (FE) simulation of hydroelastic wave-structure interaction problems. It solves a coupled problem for the linearised fluid velocity potential, the free-surface elevation, and, possibly, structural deflections of multi-body flexible structures. HydroElasticFEM.jl uses a monolithic approach to solve the coupled problem, as originally proposed in (Colomés et al., 2023). The package is built on Gridap.jl (Verdugo & Badia, 2022), a high-level Julia FE framework, and supports both structured Cartesian meshes and three-dimensional unstructured meshes imported from Gmsh (Geuzaine & Remacle, 2009).

The package provides frequency-domain and time-domain simulation modes. Frequency-domain mode solves a single complex-valued linear system for monochromatic wave excitation and is the standard tool for VLFS eigenvalue analysis and transfer-function calculations. Time-domain mode integrates the equations using the Generalised- α method, giving access to transient effects and irregular sea states. Both modes share the same physics entities and mesh infrastructure. The architecture of HydroElasticFEM.jl enables the user to switch between frequency-domain and time-domain modes by changing only a configuration object.

Statement of Need

Hydroelastic analysis of VLFS demands an open, extensible, scalable, and computationally accessible simulation tool. Existing commercial packages provide robust hydrodynamic solvers, but mostly using a boundary element method (BEM) approach, and often are closed source. By nature of the solution methodology, these frameworks are limited in terms of scalability and level of fidelity. Furthermore, available hydrodynamic analysis software treat structures as rigid bodies or rely on loosely coupled structural sub-solvers, and do not expose the governing equations as user-extensible code. On the other hand, high-fidelity tools that solve the Navier-

41 Stokes equations, such as OpenFOAM or DualSPHysics, carry computational costs orders of
42 magnitude higher than linear potential-flow models for the wave-frequency regime where VLFS
43 analysis is most relevant.

44 HydroElasticFEM.jl targets ocean engineers, offshore structure designers, and academic
45 researchers who need: (i) a physically transparent model in which the governing equations
46 appear as readable Julia code; (ii) full 2D and 3D support on both structured and unstructured
47 meshes; and (iii) a plugin architecture that allows new structural models to be added without
48 modifying the assembly core. By building on Gridap.jl, the package inherits automatic
49 differentiation, a broad finite element space library, and an active open-source community.

50 State of the Field

51 Commercial tools for potential flow-based hydrodynamic analysis, e.g. WAMIT, ANSYS Aqua,
52 ORCAWAVE, and HydroDyn (part of OpenFAST), are widely used but carry licensing restrictions
53 and, in most cases, implement staggered fluid–structure coupling. Fully viscous models such
54 as OpenFOAM or DualSPHysics for wave-structure interaction problems are fully open but are
55 impractical for parametric frequency-domain studies.

56 In the past years the authors of this package developed a formulation for the hydroelastic
57 analysis of flexible floating structures based on a monolithic FE approach (Agarwal et al.,
58 2024; Colomé et al., 2023). The numerical tests were implemented in the open-source
59 library MonolithicFEMVLFS.jl (Colomé, 2022), which demonstrated the new monolithic
60 FE formulation for a wide variety of benchmarks. HydroElasticFEM.jl extends that work in
61 four ways: (i) 2D/3D support via a dimension-generic TankDomain{D} and a GmshDomain
62 interface for external Gmsh meshes (Geuzaine & Remacle, 2009); (ii) Membrane, Euler-
63 Bernoulli, Kirchhoff–Love plate and Timoshenko beam physics under the same plugin interface;
64 (iii) point-mass resonators for locally-resonant metamaterial applications; and (iv) arbitrary
65 number of structures with arbitrary number of rotational-spring joint formulation. In addition,
66 HydroElasticFEM.jl can take advantage of differentiable programming intrinsic capabilities to
67 compute gradients using automatic differentiation, as exploited in adjoint-based optimization
68 problems such as in (El Sayed et al., 2026).

69 Software Design

70 HydroElasticFEM.jl is organized into three layered modules: Geometry, Physics, and Simulation.

71 Figure 1 summarizes the software architecture and highlights the four main design contributions
72 of HydroElasticFEM.jl.

HydroElasticFEM.jl – Software Architecture

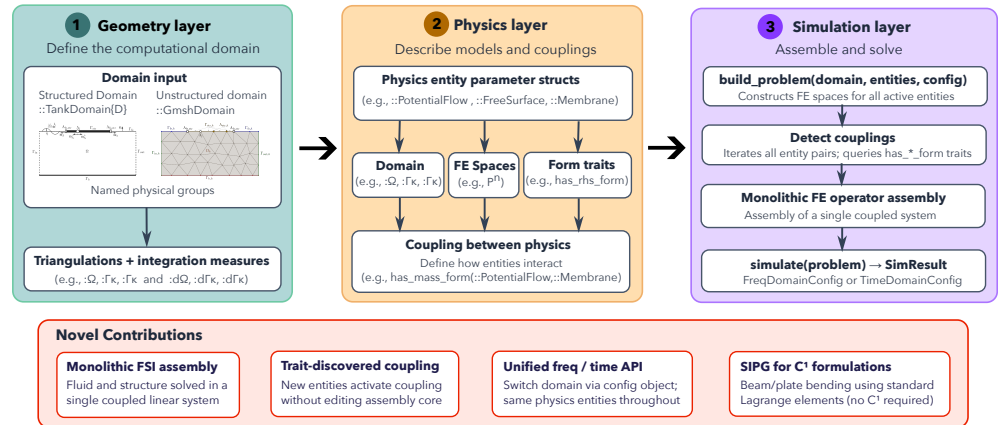


Figure 1: Software architecture of HydroElasticFEM.jl, showing the Geometry, Physics, and Simulation layers.

The **Geometry layer** converts a user-specified domain into named Gridap triangulations and quadrature measures. TankDomain{D} generates Cartesian meshes for 2D and 3D numerical wave tanks, including embedded structural sub-domains and sponge-layer damping zones. GmshDomain wraps an external .msh file where boundaries are identified by physical-group names, so the same physics code runs on any mesh that defines the required groups.

The **Physics layer** defines the weak forms of the system. Each physics entity is a parameter struct that declares its field symbols, finite element configuration, and which bilinear forms it contributes, e.g. mass, damping, stiffness, and right-hand. Cross-physics coupling (e.g., the kinematic condition at the wetted surface that couples fluid pressure to structural velocity) is declared by specializing trait functions such as has_damping_form(::PotentialFlow, ::EulerBernoulliBeam) in a dedicated coupling file. Later, the Simulation layer detects all active coupling pairs automatically by evaluating these traits over all entity combinations, so new structural models require no changes to the assembly core.

It is important to highlight that the Euler–Bernoulli beam and Kirchhoff–Love plate formulations formally require C^1 inter-element continuity, which standard Lagrange elements cannot provide. HydroElasticFEM.jl uses the Symmetric Interior Penalty Galerkin (SIPG) discontinuous Galerkin method (Colomés et al., 2023) to enforce slope continuity weakly across interior facet skeletons, achieving optimal accuracy without specialised C^1 elements.

The **Simulation layer** orchestrates four steps: building the discrete mesh model, extracting named sub-triangulations, constructing multi-field trial and test spaces for all active physics entities, and assembling the global FE operator. Users interact with two functions: build_problem, which accepts a domain, a list of physics entities, and a configuration; and simulate, which returns a SimResult from which individual solution fields are recovered by name.

Research Impact Statement

HydroElasticFEM.jl enables research on VLFS response under irregular seas, parametric design of floating breakwaters and wave energy converters with compliant elastic components, and investigation of resonator-enhanced wave attenuation in locally resonant metamaterial super-structures. The package is the principal simulation tool for hydroelastic FSI research at the Computational Multiphysics (CMOE) group at TU Delft. HydroElasticFEM.jl supports ongoing

103 projects on three-dimensional floating structures and wave energy devices within the CMOE
104 group and is the basis of several research collaborations on the topic with international partners.

105 AI Usage Disclosure

106 AI assistance was used during the development of HydroElasticFEM.jl and the preparation of
107 this paper. Specifically, Claude (Anthropic) and GitHub Copilot (Microsoft/OpenAI) were used
108 to assist with code generation, documentation drafting, and code review. All AI-generated
109 content was reviewed, verified, and modified by the authors. The authors take full responsibility
110 for the accuracy and correctness of all submitted materials.

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